Department of Information and Communication Technology

Faculty of Technology

University of Ruhuna

Object Oriented Programming Practicum-ICT 2132

Mini Project

Group Name – Strivers

Group Number – 12

FOT Portal Management Information System (MIS)

Submitted to: Mr. Nuwan Laksiri.

Group Members **:**

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1. Brief Introduction to the Problem

Managing academic and administrative activities manually in a faculty environment is inefficient and error-prone. Tasks such as:

- Managing user profiles
- Tracking attendance
- Handling marks and GPA
- Managing medical records
- Publishing notices and timetables

require a centralized system to ensure accuracy and efficiency.

2. Brief Introduction to the Solution

FOT Portal is a **client–server based desktop application** developed using:

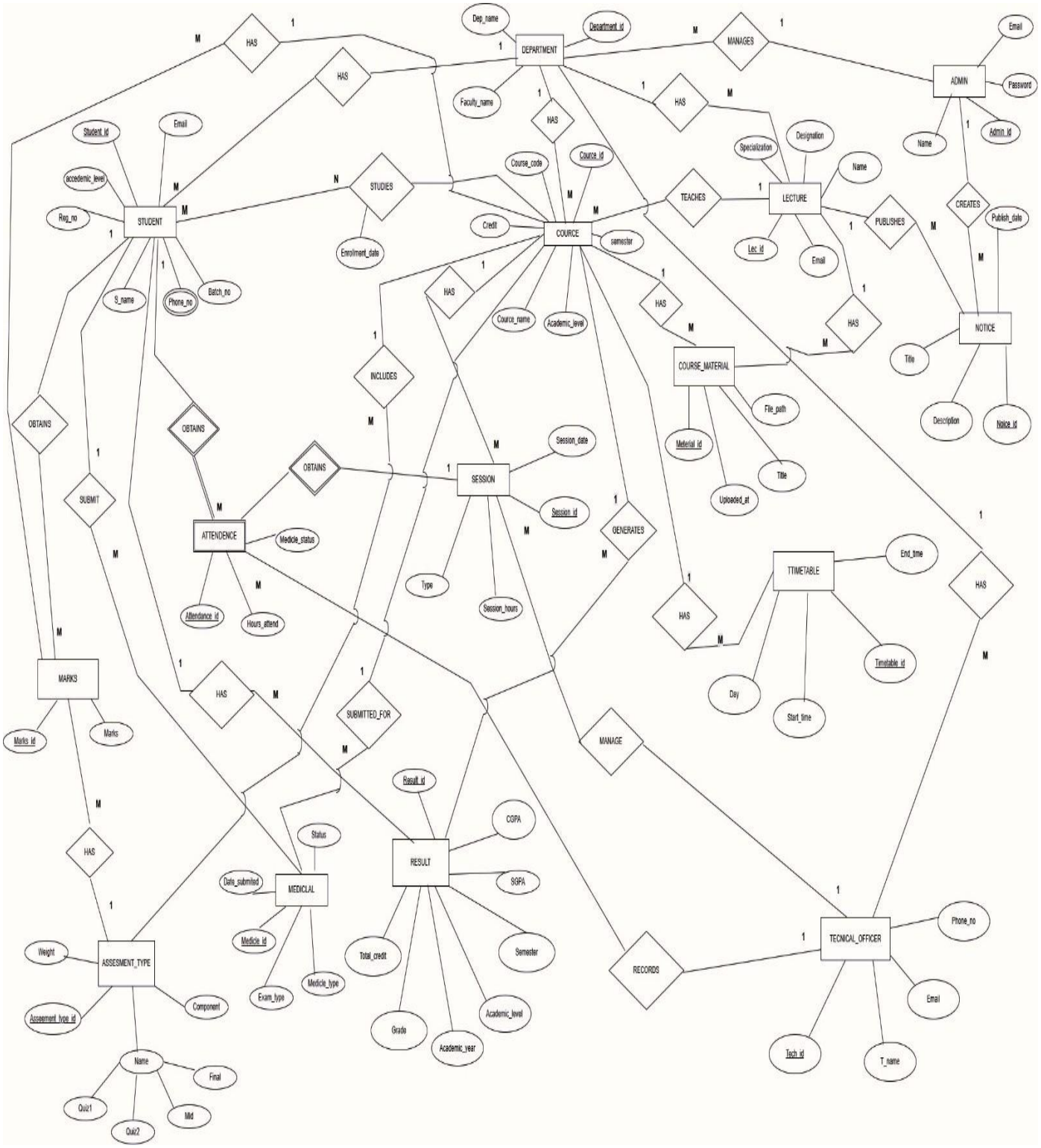
- **JavaFX (Frontend)**
- **Java Socket Programming (Backend)**
- **MySQL Database**

The system supports four user roles:

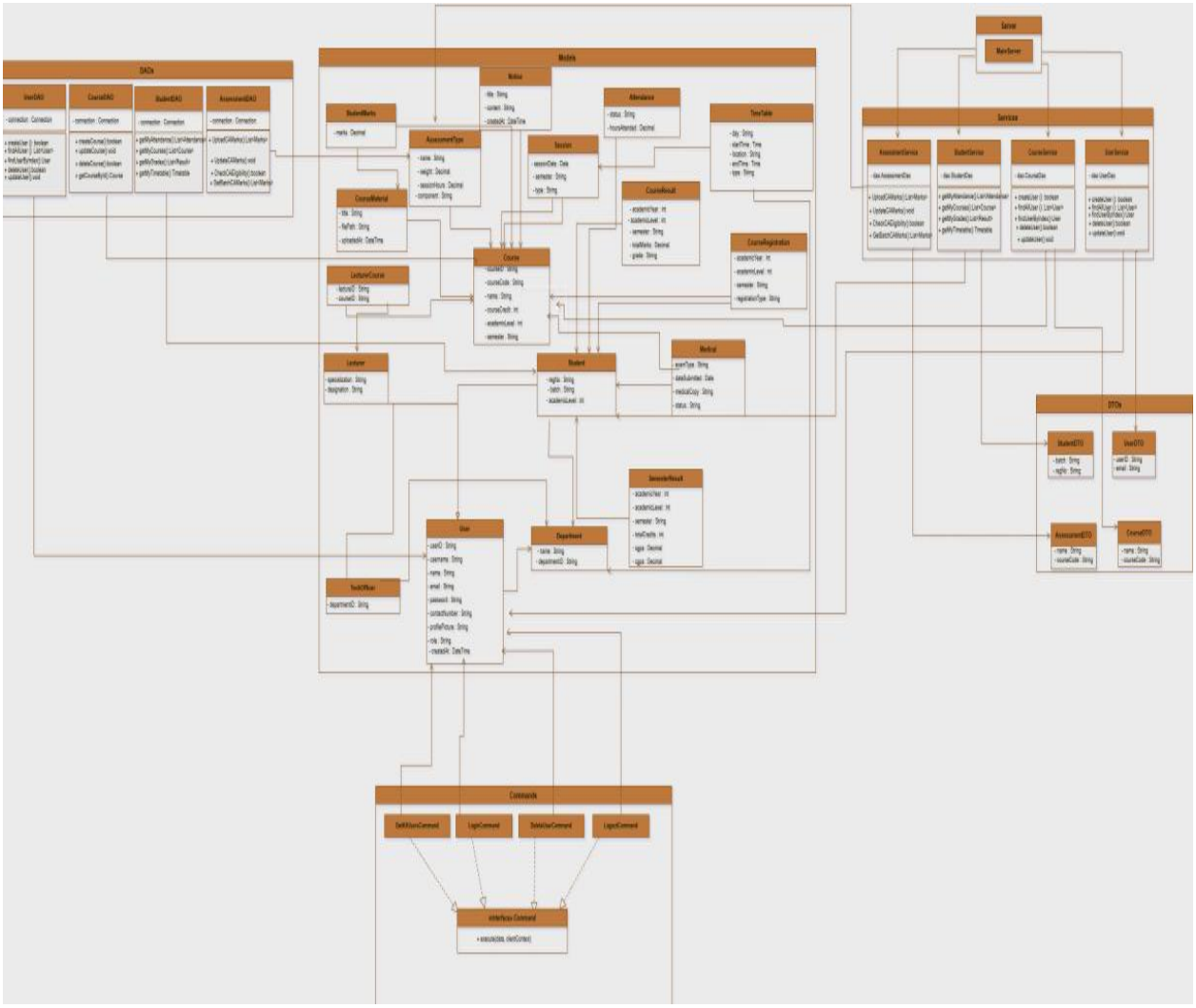
✓ Administrator
✓ Lecturer
✓ Technical Officer
✓ Undergraduate

Authentication is handled using **JWT tokens**, ensuring secure communication between client and server.

3. ER diagram for the FOT Portal Management Information System (MIS)



Class diagram for FOT Portal Management Information System (MIS)



5. Database tables used in the FOT Portal MIS

The system uses a MySQL database to store and manage all information, including user profiles, course details, attendance records, medical submissions, marks, timetables, and notices. We designed the database with connected tables using primary and foreign keys to keep the data accurate, consistent, and easy to manage.

Users Table

user_id	username	email	password	contact_number	profile_picture	role	created_at
U001	Nimal Perera	nimal@gmail.com	1234	0771234567	"C:\Users\Nuwangi\Desktop\New folder (2)\st.jpg"	Student	2026-04-26 12:18:57
U002	Kamal Silva	kamal@gmail.com	1234	0779876543	NULL	Student	2026-04-26 12:18:57
U003	Dr. Fernando	fernando@gmail.com	1234	0712345678	C:\Users\Nuwangi\Desktop\New folder (2)\st.jpg	Lecturer	2026-04-26 12:18:57
U004	Ms. Perera	perera@gmail.com	1234	0723456789	C:\Users\Nuwangi\Desktop\New folder (2)\lec.jpg	Lecturer	2026-04-26 12:18:57
U005	Admin User	admin@gmail.com	admin123	0700000000	"C:\Users\Nuwangi\Desktop\New folder (2)\admin.jpg"	Admin	2026-04-26 12:18:57
U006	Tech Officer 1	tech@gmail.com	1234	0751234567	C:\Users\Nuwangi\Desktop\New folder (2)\admin.jpg	Tech_Officer	2026-04-26 12:18:57

Students Table

user_id	reg_no	batch	academic_level	department_id
U001	ICT2023001	2023	1	ICT
U002	ICT2023002	2023	1	ICT

Lecturers Table

user_id	specialization	designation
U003	Software Engineering	Senior Lecturer
U004	Database Systems	Lecturer

Tech Officers Table

user_id	department_id
U006	ICT

Courses Table

course_id	course_code	name	course_credit	academic_level	semester	department_id
C001	ICT2122	Object Oriented Programming	2	2	1	ICT
C002	ICT2113	Data Structure and Algorithm	3	2	1	ICT

Course Registration Table

student_id	course_id	academic_year	semester	registration_type
U001	C001	2024	1	Proper
U001	C002	2024	1	Proper
U002	C001	2024	1	Proper
U002	C002	2024	1	Proper

Assessment Types Table

assessment_type_id	course_id	name	weight	component
1	C001	Quiz 1	10.00	CA
2	C001	Mid Exam	30.00	CA
3	C001	Final Exam	60.00	Final
4	C002	Quiz 1	10.00	CA
5	C002	Mid Exam	30.00	CA
6	C002	Final Exam	60.00	Final
7	C002	Final	100.00	Final
8	C001	Final	100.00	Final

Student Marks Table

mark_id	student_id	assessment_type_id	marks
1	U001	1	75.00
2	U001	2	65.00
3	U001	3	80.00
4	U002	1	60.00
5	U002	2	70.00
6	U002	3	75.00
7	U001	7	85.00
8	U001	8	98.00
9	U002	7	56.00
10	U002	8	96.00

Attendance Table

attendance_id	student_id	session_id	status	hours_attended
1	U001	1	Present	2.00
2	U001	2	Absent	0.00
3	U002	1	Present	2.00
4	U002	2	Present	2.00
5	U001	3	Present	2.00
9	U002	3	Present	2.00

Medical Table

medical_id	student_id	course_id	exam_type	date_submitted	medical_copy	status
1	U001	C001	Mid	2026-04-26	C:\Users\Nuwangi\Downloads\medical.pdf	Pending
2	U002	C001	Final	2026-04-24	C:\Users\Nuwangi\Downloads\medical.pdf	Pending
3	U001	C001	Attendance	2024-01-12	C:\Users\Nuwangi\Downloads\medical.pdf	Pending

Notice Table

notice_id	title	description	pdf_file_path	created_by	created_at
1	Semester Start	Semester will start on Feb 1	C:\Users\Nuwangi\Desktop\Document 1.pdf	U005	2026-04-26 12:49:24
2	Semester Start	Semester will start on Feb 1	notice1.pdf	U005	2026-04-26 14:17:11
3	Exam Notice	Mid exams schedule released	notice2.pdf	U005	2026-04-26 14:17:11

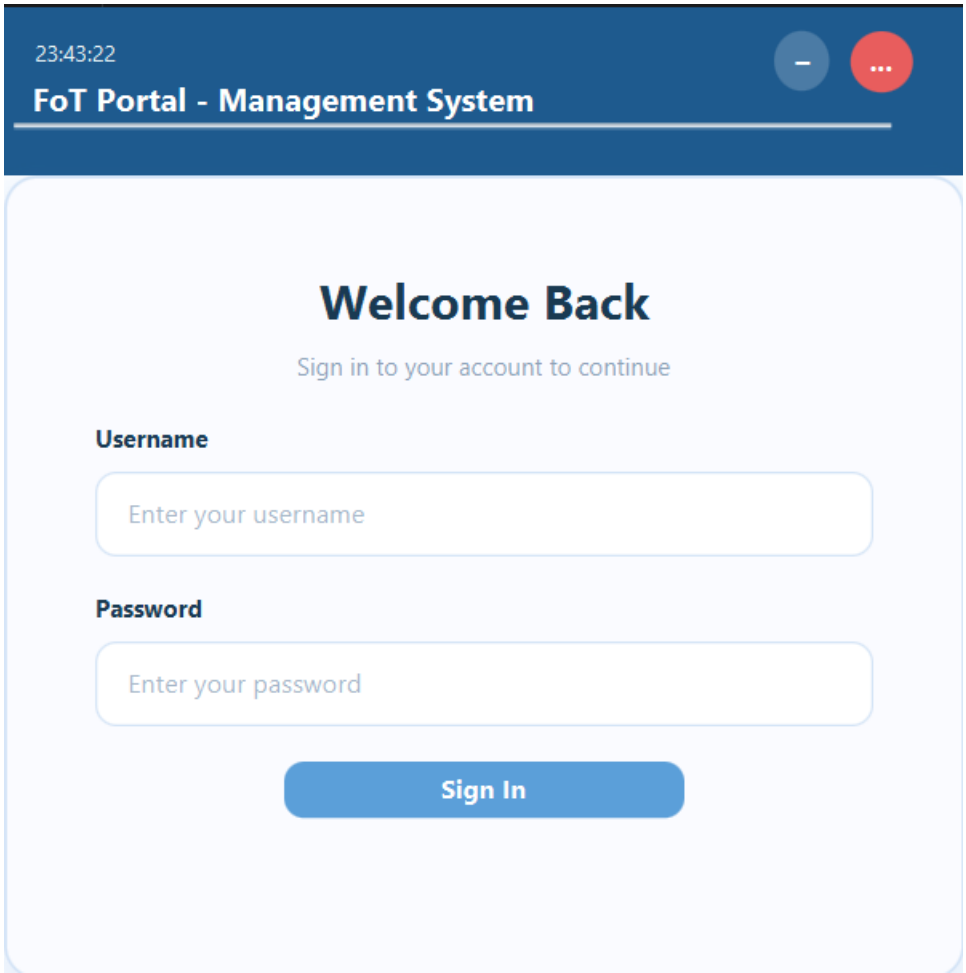
Timetable Table

timetable_id	department_id	academic_level	semester	title	pdf_file_path	uploaded_at
1	ICT	3	1	Level 1 Semester 1 Timetable	C:\Users\Nuwangi\Downloads\Document 1.pdf	2026-04-26 12:47:34
2	ICT	1	1	Level 1 Semester 1 Timetable	tt1.pdf	2026-04-26 14:17:28

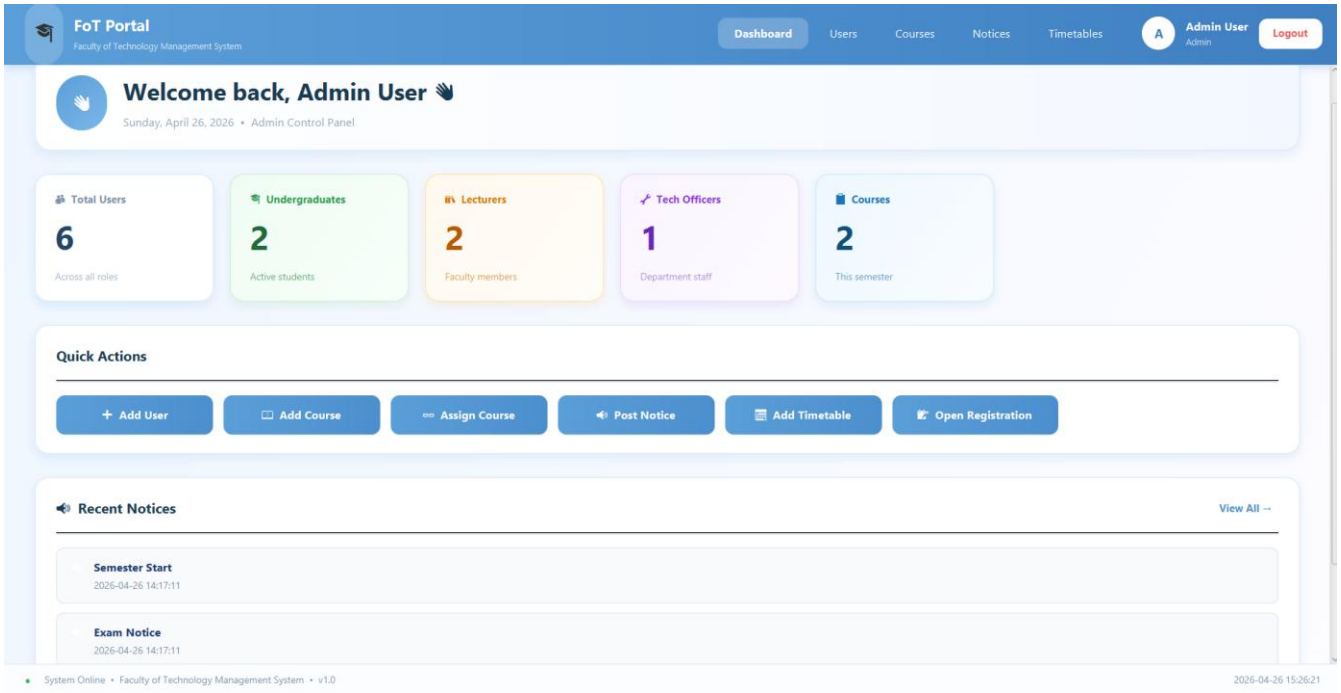
Additionally, many to many relationship tables are included in MIS database. If users update their details then database is automatically updated.

6. User Interfaces Design in Management Information System (MIS).

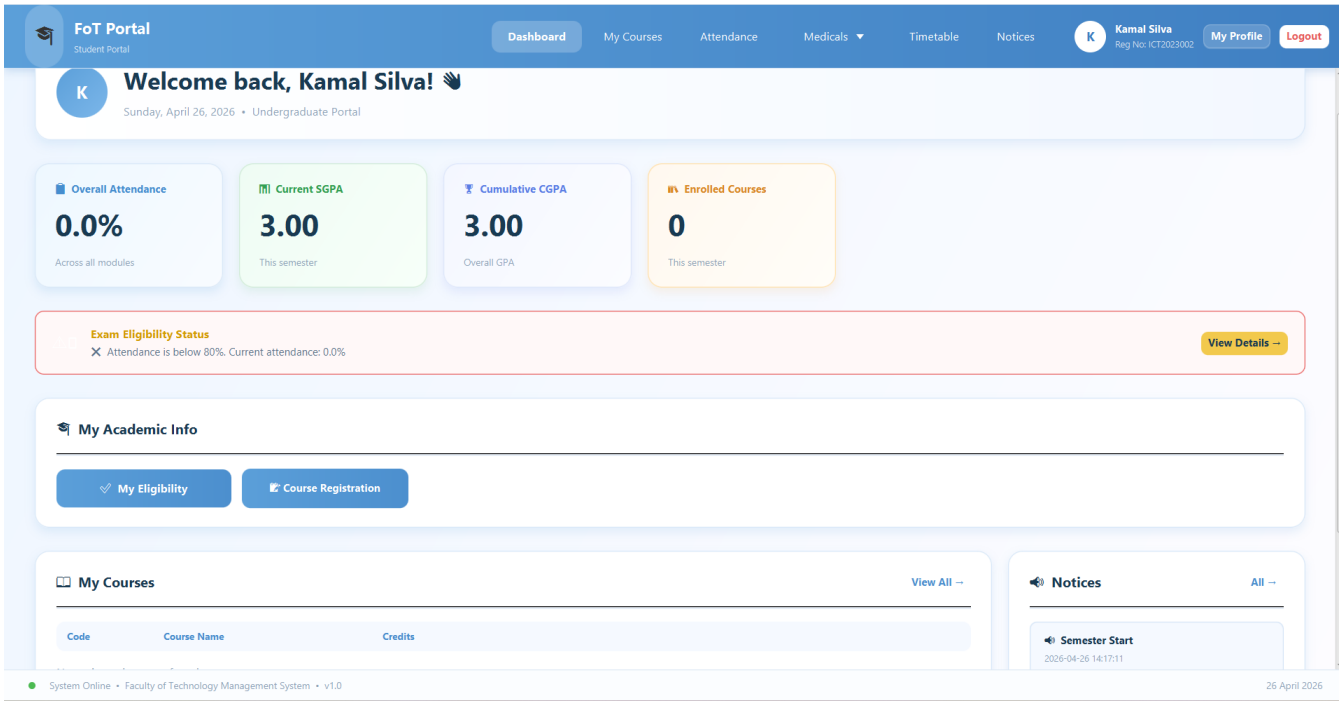
Login Page



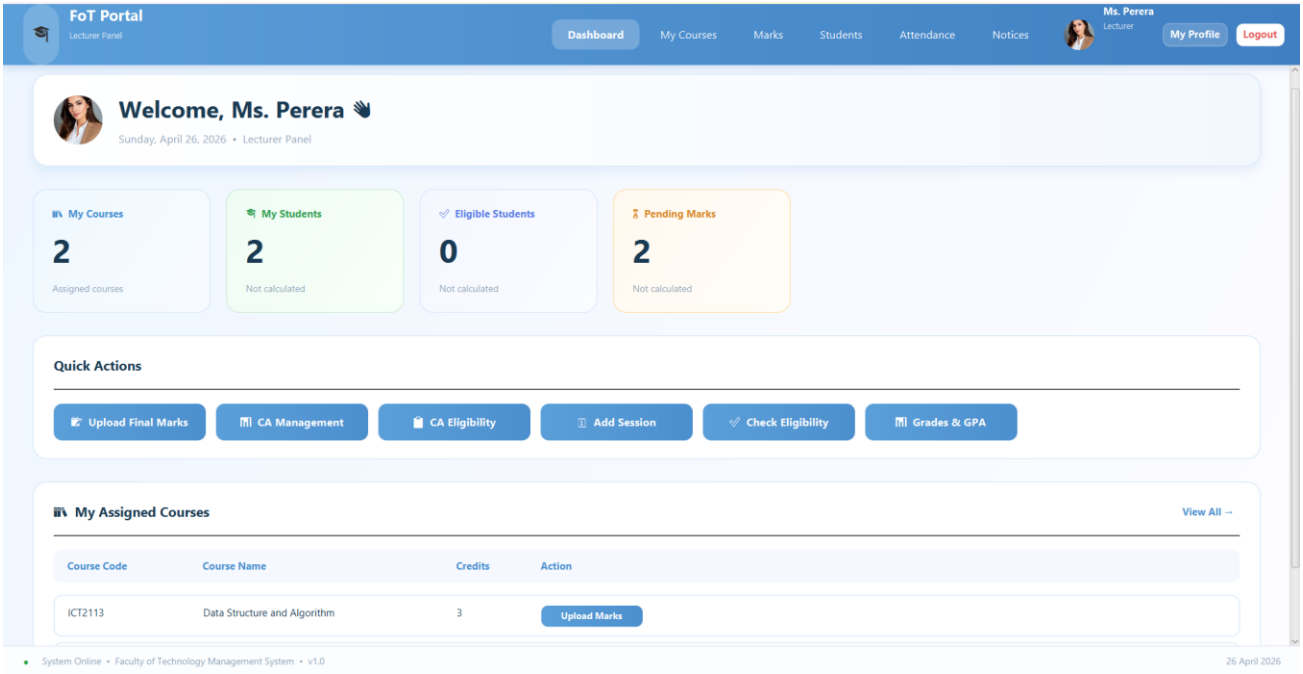
Admin Dashboard



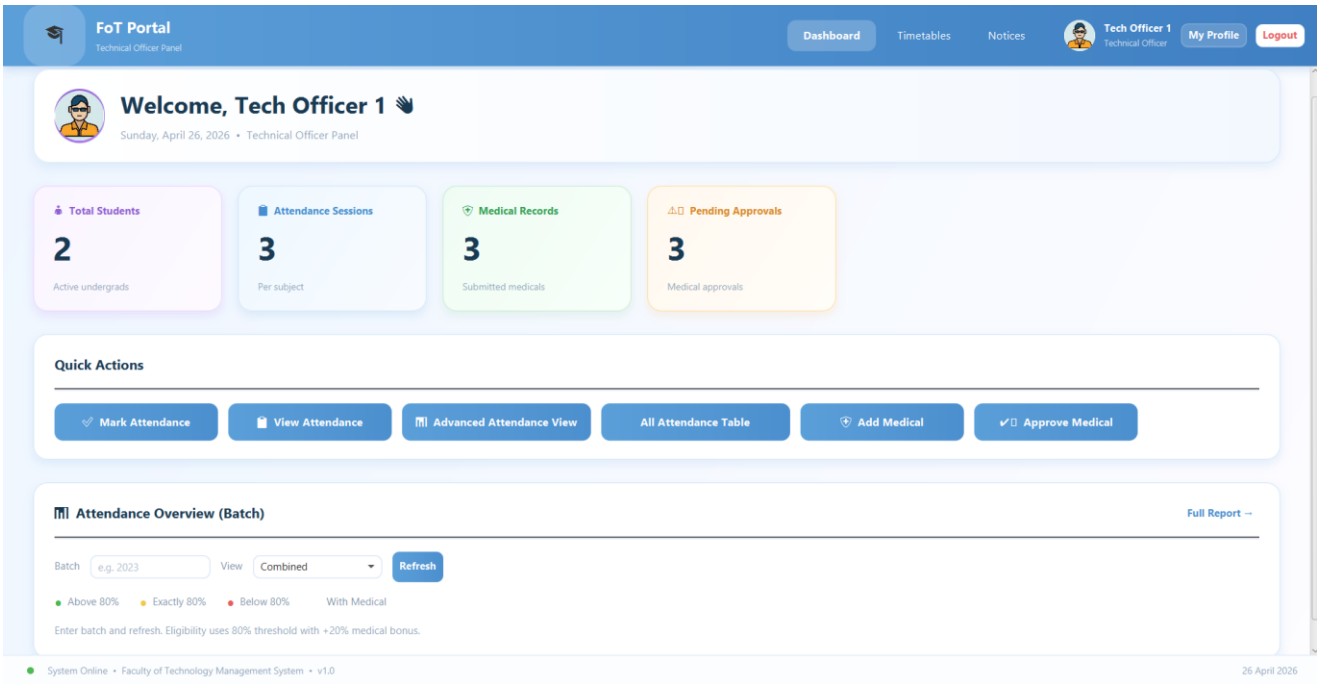
Student Dashboard



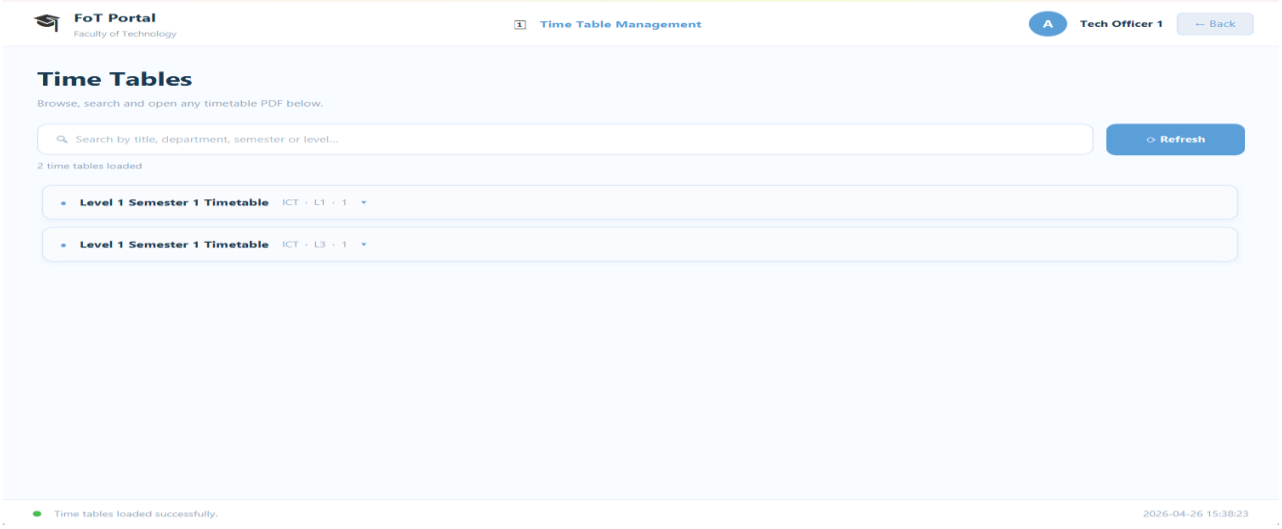
Lecturer Dashboard



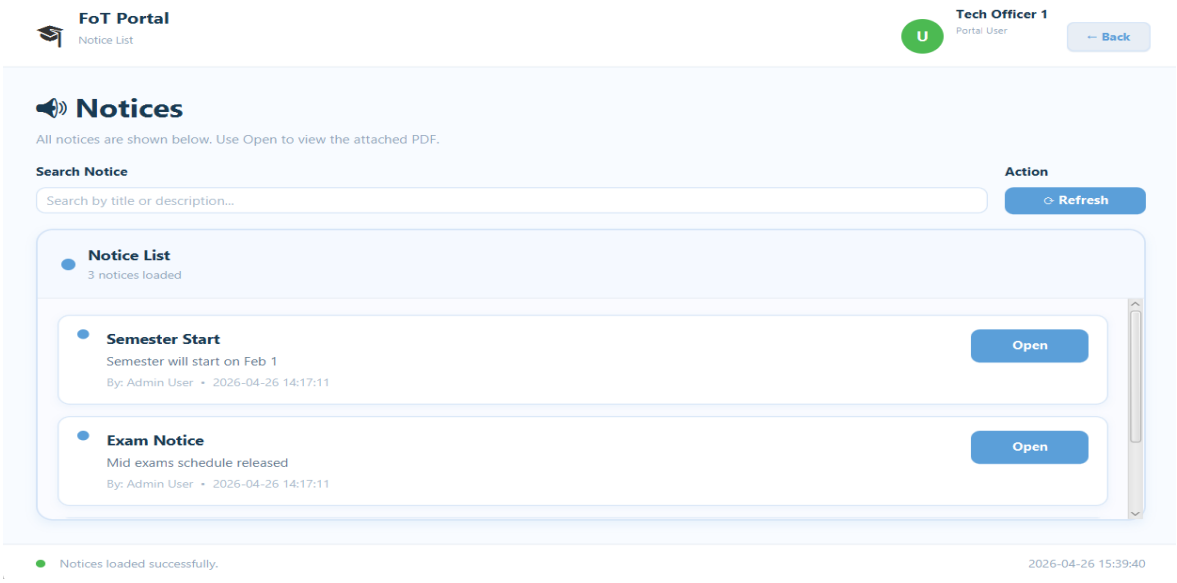
Technical officer



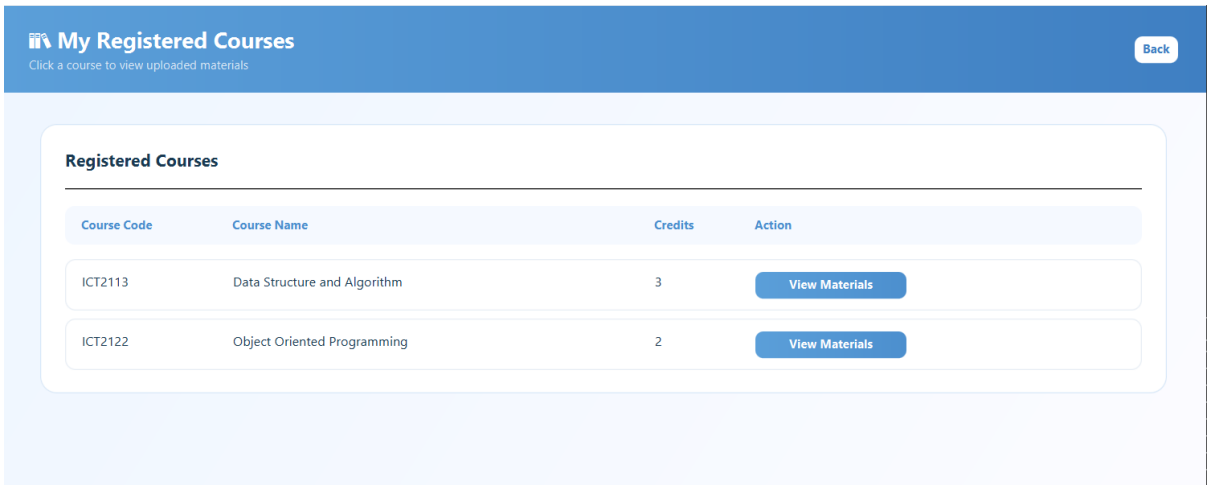
Time table



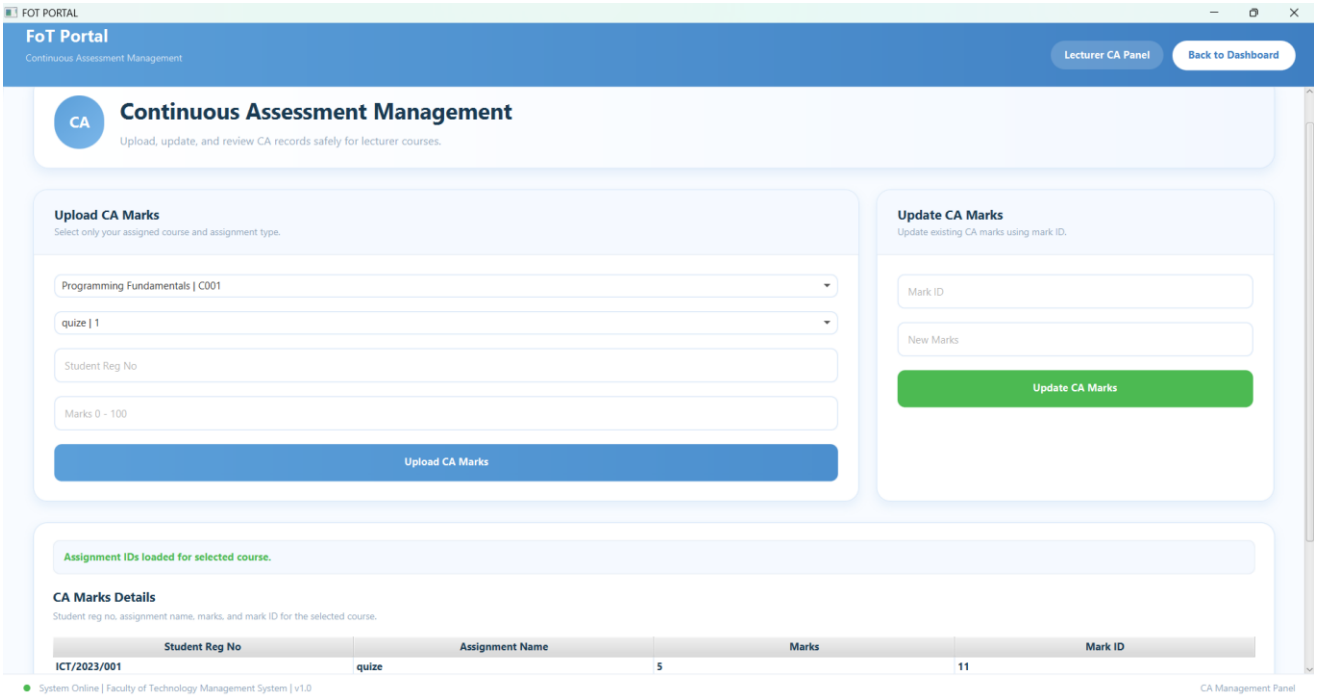
Notice



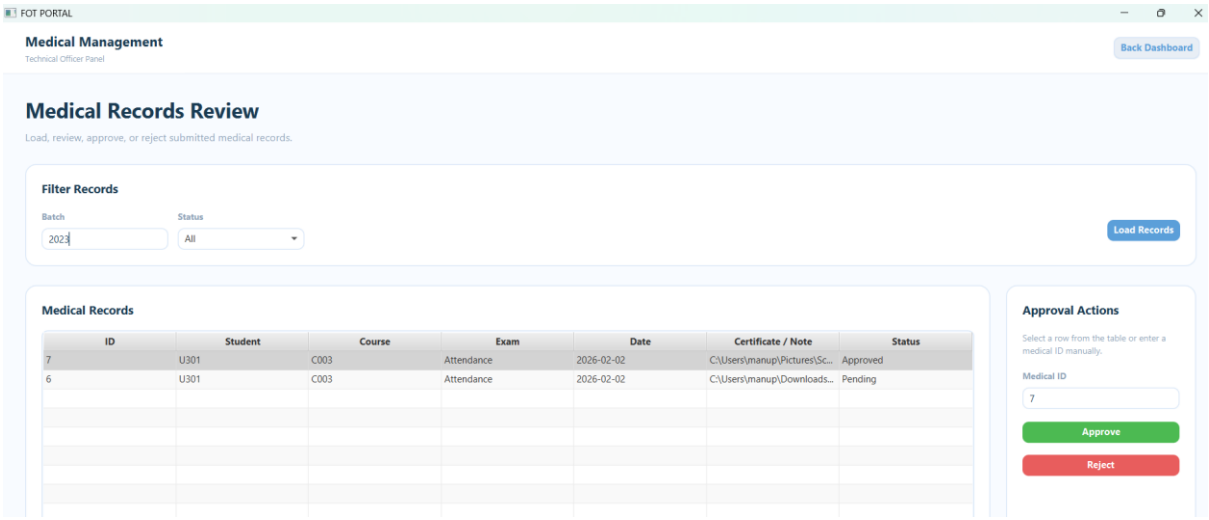
Courses



CA Management



Medical Management



7. User Roles and Justification for each Role

Admin Management Requirements

- Manage users
- Manage courses
- Publish notices
- Manage timetables
- Approve medicals

Lecturer Management requirements

• Lecturer can update their profile except username and password.

• Lecturer can modify and add materials to courses.
• Lecturer can upload marks for all kind of exams.
• Lecturer can see student details.
• Lecturer can see student eligibility.
• Lecturer can see student marks, grades, and GPA.
• Lecturer can see attendance and medical records of students.
• Lecturer can see notices.

Technical Officer Management requirements

• Can add and maintain attendance details of students.

• Can add and maintain medical details of students.
• Can see notices.
• Can see timetables of their department

Undergraduate Management requirements

• Can update only contact details and profile picture of their profile.

• Can see attendance details. • Can see medical details.
• Can see their course details.
• Can see their grades and GPA.
• Can see their timetables.
• Can see notices.

8. Tools and technologies used in the project

• IntelliJ idea and javafx (scene builder)

To create and maintain GUI (Graphical User Interface)

To maintain java codes

To Connect the database using MySQL connectors

• MYSQL server

To create and manage database efficiency

To create table structures and data insertion

• Draw.io

Used to create ER diagram and Class diagram

• GitHub and GitHub Desktop

Served as the version control platform for tracking source code changes, collaborating within the team, and managing different versions of the system throughout development.

• Microsoft Word

Used in documentation processes, such as preparing the Final Project Report and the Software Requirements Specification (SRS) document.

9. Security Measures Implemented in the system.

- Role-Based Access Control (RBAC)

Access to system functionalities is granted based on the user's role (Admin, Lecturer, Technical Officer, or Undergraduate). Each user is directed to their respective dashboard after login, restricting unauthorized access to sensitive modules.

- Credential Verification with Database Validation

Login credentials are securely verified in the MySQL database, preventing unauthorized access and ensuring only valid users can enter the system.

- Limited UI Access Based on User Type

Each interface is dynamically rendered based on the logged-in user's role, ensuring users can only interact with features relevant to their role, thereby minimizing exposure to unauthorized data.

- Data Validation and Input Sanitization

Input fields across the system are validated to prevent malformed entries and possible SQL injection attacks. This ensures data consistency and security.

- Secure Database Connectivity

The system uses secure and efficient MySQL connectors in Java to manage database interactions, reducing risks related to data leakage or interception.

- Version Control and Backup Using GitHub

Project code is maintained under GitHub with frequent commits, allowing version history tracking and restoring backups in case of system failure or accidental data loss.

10.Challenges Faced during the Development

• **Database complexity:** Mapping relationships among multiple entities (e.g., medical, attendance) was challenging.

• **GUI synchronization**: Ensuring that the user interface reflects real-time updates from the database .

• **Team coordination:** Dividing tasks and merging contributions without conflicts was time-consuming.

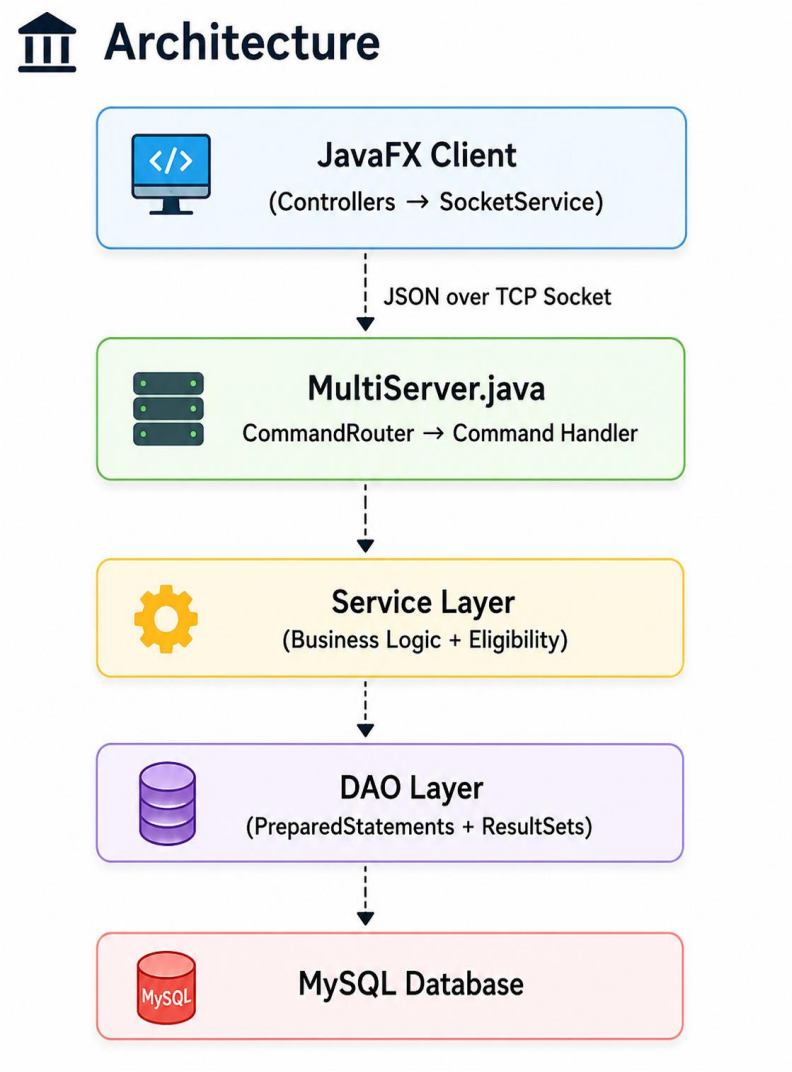
11. Solutions to Overcome Identified Challenges

• **Database complexity:** We created a detailed ER diagram to represent FOT Portal – Faculty Management System details.

• **GUI Synchronization:** We used a single main frame with multiple panels, switching panels inside the main frame based on user actions to keep the interface simple and organized.

• **Team coordination:** Used GitHub for version control, assigned tasks to project, and conducted regular group communication.

12. System Architecture



13.Individual contribution to System development

TG_no	Name	Individual contribution
TG/2023/1773	K.G.M.R.BANDARA	Tech officer – Mark Attendance – View Attendance – Medical Management – Attendance Overall view – Tech officer Dashboard – Tech officer Profile Student – Medical section Lecturer – CA Managment – CA Eligibility Class diagram designing
TG/2023/1727	D.S.J. DE SILVA	– My Course Management – Marks management – Grades and GPA generating – Lecture dashboard – ER Diagram designing – Student view – Attendance that lecturer's course following student only. – Generating Report GPA and Marks pdf

TG/2023/1756	M.I.M. IRFAN	Admin – User management – Course management – Add notice – Add Time Table – Course Registration – Admin dashboard Lecturer – Add Session – Check eligibility – Profile section update Student – Course Registration – Profile section update System Designing
TG/2023/1736	U.G.A.N.KAUSHALYA	Student • My Course -View Courses -View Course material • Attendance View • Time table View • Notice View • My eligibility checking • Students Dashboard for a student -Overall Attendence -Current GPA -Calculate CGPA -Enrolled Courses • ER diagram designing

14. GitHub Repository Link

https://github.com/Manupriya144/JavaMiniProject.git

15.References

- Java oracle documentation
- Javafx documentation
- GitHub documentation
- Ai assistant (chatgpt)
- Ai assistant (claude)